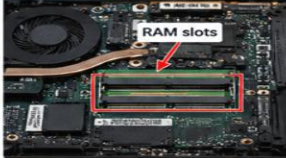


Module-4 (Assignment Day 4)

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.



1. Shut down and unplug  Shut down the computer and unplug the power cable.	2. Open the case and locate RAM  Open the cabinet/back cover and locate the RAM slots on the motherboard.	3. Remove the RAM module  Release the side clips on the RAM slot and carefully remove the RAM module.	4. Check for dust or damage  Check the RAM module and slot for dust, damage, or loose connections.	
5. Align the notch  Align the notch on the RAM with the slot and place it correctly.	6. Press down until locked  Press the RAM down firmly until both side clips lock into place.	7. Close the case and reconnect  Close the cabinet/back cover and reconnect the power cable.	8. Power on the computer  Power on the computer and check whether it starts normally.	9. Verify RAM in BIOS/Task Manager  Verify the RAM in BIOS or Windows Task Manager to confirm it is detected.
10. Result and Issues				
 Result No physical damage was found, and the computer booted normally after reseating the RAM.		 Issue Observed No issues were encountered during the process. The RAM module was properly seated, and the system recognized it without any errors.		

RAM Reseating – Step-by-Step Documentation

1. Shut down the computer and unplug the power cable.
2. Open the cabinet/back cover and locate the RAM slots on the motherboard.
3. Release the side clips on the RAM slot and carefully remove the RAM module.
4. Check the RAM module and slot for dust, damage, or loose connections.
5. Align the notch on the RAM with the slot and place it correctly.
6. Press the RAM down firmly until both side clips lock into place.
7. Close the cabinet/back cover and reconnect the power cable.
8. Power on the computer and check whether it starts normally.

9. Verify the RAM in BIOS or Windows Task Manager to confirm it is detected.
10. Issue observed: No physical damage was found, and the computer booted normally after reseating the RAM.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

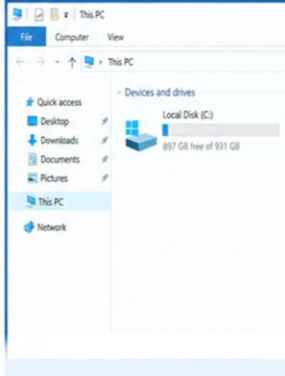


GPU Troubleshooting Checklist :-

1. **Identify the symptom** – Check for no display, screen flickering, artifacts, crashes, or poor gaming performance.
2. **Check the GPU connection** – Turn off the PC and make sure the GPU is properly seated in the PCIe slot.
3. **Check power cables** – Ensure the GPU's 6/8-pin PCIe power connectors are firmly connected.
4. **Check the display cable** – Make sure the HDMI/DisplayPort cable is connected to the GPU, not the motherboard.
5. **Check GPU drivers** – Open Device Manager and update or reinstall the graphics driver if there is an error.
6. **Check GPU temperature** – Use a monitoring tool to check for overheating during gaming.
7. **Test the GPU** – Run a game or graphics benchmark and check for crashes, artifacts, or display problems.
8. **Confirm the fix** – Restart the PC and verify that the display works normally and games run without GPU-related errors.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.



1 Power off and unplug the PC	2 Disconnect SATA data cable	3 Drive not detected in BIOS	4 Boot error message
			
Power off the PC and unplug the power cable before handling the drive.	Disconnect the SATA data cable from the SSD/HDD to simulate a storage connection problem.	Power on the PC and enter the BIOS/UEFI setup. The storage drive was not detected in the SATA/Storage Information section.	The system may also show an error such as "Boot Device Not Found" or "No bootable device."
5 Reconnect SATA cables	6 Drive detected in BIOS	7 Boot into Windows	8 Result
			
Power off the PC again and reconnect the SATA data cable firmly. Also check the SATA power connector.	Power on and enter BIOS again. The SSD/HDD now appears correctly with its model and storage capacity.	Boot into Windows and confirm that the operating system and files are accessible.	The problem was caused by the disconnected SATA cable. After reconnecting it, the drive was detected and the system booted normally.

Storage Issue Simulation – SATA SSD/HDD

1. **Power off the PC** and unplug the power cable before handling the drive.
2. **Disconnect the SATA data cable** from the SSD/HDD to simulate a storage connection problem.

3. **Power on the PC** and enter the BIOS/UEFI setup. The storage drive was **not detected** in the SATA/Storage Information section.
4. The system may also show an error such as “**Boot Device Not Found**” or “**No bootable device.**”
5. **Power off the PC** again and reconnect the SATA data cable firmly. Also check the SATA power connector.
6. **Power on and enter BIOS** again. The SSD/HDD now appears correctly with its model and storage capacity.
7. **Boot into Windows** and confirm that the operating system and files are accessible.
8. **Result:** The problem was caused by the disconnected SATA cable. After reconnecting it, the drive was detected and the system booted normally.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.



Component I would check first: RAM

Faulty or poorly seated RAM can cause system instability, crashes, and random restarts. It is also easy to check and reseat.

Diagnostic Steps for RAM

1. Shut down the PC and unplug the power cable.
2. Remove and reseat the RAM firmly in the correct motherboard slot.
3. Check for dust or physical damage on the RAM and slot.
4. Test one RAM module at a time if the PC has multiple modules.
5. Run Windows Memory Diagnostic:
6. If errors are found, test the RAM in another compatible slot or use a known-good RAM module.
7. Use the PC normally or run a memory-intensive application to check whether the random restarts continue.